



COMPUTER

Topic: Processes

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01. **A process is best defined as:**
 - (a) A program stored on disk
 - (b) A program in execution
 - (c) A compiled code file
 - (d) A thread inside CPU
02. **Which of the following is NOT part of a process?**
 - (a) Program counter
 - (b) CPU registers
 - (c) Source code file
 - (d) Stack
03. **The PCB is used to store:**
 - (a) Program instructions
 - (b) Process management information
 - (c) Only memory mapping
 - (d) Only scheduling data
04. **Which state indicates the process is waiting for CPU?**
 - (a) Running
 - (b) Waiting
 - (c) Ready
 - (d) Terminated
05. **Which state indicates waiting for I/O completion?**
 - (a) Ready
 - (b) Running
 - (c) Waiting/Blocked
 - (d) New
06. **Context switching occurs when:**
 - (a) Process terminates
 - (b) CPU switches between processes
 - (c) I/O operation starts
 - (d) Interrupt disabled
07. **Which of these causes a context switch?**
 - (a) Time quantum expiry
 - (b) Power failure
 - (c) Disk formatting
 - (d) Compiler error
08. **Which is stored in PCB for resuming execution?**
 - (a) Cache memory
 - (b) Program counter
 - (c) Disk block number
 - (d) Page fault rate
09. **Process ID (PID) is used for:**
 - (a) Memory allocation
 - (b) Unique process identification
 - (c) Scheduling priority
 - (d) Deadlock handling
10. **Which scheduler selects next process for CPU?**
 - (a) Long-term
 - (b) Medium-term
 - (c) Short-term
 - (d) I/O scheduler
11. **Long-term scheduler controls:**
 - (a) CPU speed
 - (b) Degree of multiprogramming
 - (c) Context switching
 - (d) I/O operations
12. **Medium-term scheduler performs:**
 - (a) Process creation
 - (b) Swapping
 - (c) Scheduling algorithm selection
 - (d) Interrupt handling
13. **Ready queue contains:**
 - (a) All terminated processes
 - (b) Processes waiting for CPU
 - (c) Processes doing I/O
 - (d) Zombie processes
14. **Device queue contains:**
 - (a) Ready processes
 - (b) Running processes
 - (c) Processes waiting for I/O
 - (d) Suspended processes
15. **Which transition is valid?**
 - (a) Running → New
 - (b) Waiting → Running
 - (c) Running → Ready
 - (d) Terminated → Ready
16. **Which transition occurs after I/O completion?**
 - (a) Waiting → Ready
 - (b) Running → Waiting
 - (c) Ready → Running
 - (d) New → Ready
17. **Which is NOT stored in PCB?**
 - (a) CPU registers
 - (b) Scheduling info
 - (c) Program binary
 - (d) Process state
18. **Which process is likely I/O-bound?**
 - (a) Matrix multiplication
 - (b) File download
 - (c) Encryption
 - (d) Sorting algorithm
19. **CPU-bound processes spend most time in:**
 - (a) Waiting
 - (b) Ready
 - (c) Running
 - (d) Terminated
20. **Context switch overhead means:**
 - (a) Faster execution
 - (b) CPU idle work
 - (c) Extra OS work
 - (d) Reduced memory usage
21. **Stack segment stores:**
 - (a) Global variables
 - (b) Dynamic memory
 - (c) Function calls
 - (d) Page tables
22. **Heap segment stores:**
 - (a) Local variables
 - (b) Dynamic memory
 - (c) Registers
 - (d) Instructions
23. **Which state is initial?**
 - (a) Ready
 - (b) Running
 - (c) New
 - (d) Waiting

24. **Which state is final?**
 (a) Waiting (b) Ready
 (c) Terminated (d) Running
25. **Process creation involves:**
 (a) Deleting PCB (b) Assigning PID
 (c) Killing parent (d) Swapping
26. **Which event moves Ready → Running?**
 (a) I/O completion (b) CPU allocation
 (c) Interrupt (d) Timeout
27. **Which event moves Running → Waiting?**
 (a) Quantum expiry (b) I/O request
 (c) Termination (d) Swap out
28. **Zombie process means:**
 (a) Running process (b) Finished but PCB exists
 (c) Waiting for CPU (d) Suspended
29. **Orphan process occurs when:**
 (a) Child dies (b) Parent terminates
 (c) CPU fails (d) Scheduler stops
30. **Which IPC method is faster?**
 (a) Message passing (b) Shared memory
 (c) Signals (d) Pipes
31. **Message passing is:**
 (a) Faster than shared memory
 (b) Requires synchronization primitives
 (c) Safer communication
 (d) Kernel independent
32. **Process state stored for switching:**
 (a) Cache (b) PCB
 (c) Disk (d) RAM only
33. **Which causes Running → Ready?**
 (a) I/O request (b) Time slice expiry
 (c) Termination (d) Swap in
34. **Multiprogramming aims to:**
 (a) Increase idle time
 (b) Maximize CPU utilization
 (c) Reduce memory
 (d) Eliminate waiting
35. **Which is true about processes?**
 (a) Share memory always
 (b) Independent address space
 (c) Cheaper than threads
 (d) No OS involvement
36. **Which is heavier?**
 (a) Thread (b) Process
 (c) Register (d) Signal
37. **Which is NOT a scheduler type?**
 (a) Long-term (b) Short-term
 (c) Medium-term (d) Priority-term
38. **Process termination releases:**
 (a) CPU registers only (b) Memory & resources
 (c) PCB only (d) Stack only
39. **Which info helps scheduling decision?**
 (a) PC (b) Priority
 (c) Stack pointer (d) Heap size
40. **Degree of multiprogramming =**
 (a) Number of CPUs
 (b) Processes in ready queue
 (c) Processes in memory
 (d) Threads per process
41. **Which state consumes CPU?**
 (a) Ready (b) Waiting
 (c) Running (d) New
42. **Dispatcher is responsible for:**
 (a) Creating processes (b) Context switching
 (c) Swapping (d) Scheduling policy
43. **Which is NOT an IPC example?**
 (a) Shared memory (b) Message queue
 (c) Registers (d) Pipe
44. **Process switching requires saving:**
 (a) Hard disk state (b) CPU registers
 (c) Compiler output (d) BIOS data
45. **Which is correct?**
 (a) Ready → Waiting directly
 (b) Running → Waiting
 (c) Waiting → Running directly
 (d) Terminated → Running
46. **Which improves responsiveness?**
 (a) Long jobs first (b) Time-sharing
 (c) Batch OS (d) FCFS only
47. **Which process likely CPU-bound?**
 (a) Video streaming
 (b) OS update download
 (c) Scientific simulation
 (d) Keyboard input
48. **Which structure links processes in queues?**
 (a) Stack (b) PCB pointers
 (c) Cache (d) Heap
49. **Which state after admission?**
 (a) Running (b) Ready
 (c) Waiting (d) Terminated
50. **Main goal of scheduling:**
 (a) Increase waiting time
 (b) Fair & efficient CPU use
 (c) Reduce memory
 (d) Avoid interrupts